

Question 01

```
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>javac Assignment_02.java

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>java Q_01
Name - Githmi Thattsarani
Address - Near the 4th mile post Bombuwala
Phone Number - 8762395834

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>
```

```
File Edit Format View Help
class Q_01{
    public static void main (String [] args){
        System.out.println("Name - " + "Githmi Thathsarani");
        System.out.println("Address - " + "Near the 4th mile post Bombuwala");
        System.out.println("Phone Number - " + "0762395034");
    }
}
```

Question 02

```
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>javac Assignment
t_02.java

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>java Q_02
*
**
***
****
*****
*****
*****

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>
```

```
class Q_02{  
    public static void main (String [] args){  
        System.out.println("*");  
        System.out.println("****");  
        System.out.println("*****");  
        System.out.println("*****");  
        System.out.println("*****");  
        System.out.println("*****");  
        System.out.println("*****");  
    }  
}
```

Question 03

```
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>javac Ass.
t_02.java

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>java Q_03
299
+ 800
-----
1099
=====

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>
```

```
File Edit Format View Help
class Q_03{
    public static void main (String [] args){
        System.out.println("          299");
        System.out.println("+ "      " + "800");
        System.out.println("-----");
        System.out.println("          1099");
        System.out.println("=====");
    }
}
```

Question 04

[illegible][illegible]

Question 05

Program A

```
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>javac Assignment_02.java

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>java Q_05pA
Institute of Computer Engineering Technology
223 A,
Galle Road,
Panadura.

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>_
```

```
File Edit Format View Help
class Q_05pA{
    public static void main(String args[]){
        System.out.println("Institute of Computer Engineering Technology");
        System.out.println("223 A.");
        System.out.println("Galle Road.");
        System.out.println("Panadura.");
    }
}
```

Program B

```
Panadura.

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>javac Assignment_02.java

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>java Q_05pB
Institute of Computer Engineering Technology
223 A,Galle Road,Panadura.
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>
```

```
File Edit Format View Help
class Q_05pB{
    public static void main(String args[]){
        System.out.println("Institute of Computer Engineering Technology");
        System.out.println("223 A.");
        System.out.println("Galle Road.");
        System.out.println("Panadura.");
    }
}
```

Question 06

```
Institute of Computer Engineering Technology
223 A,Galle Road,Panadura.
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>javac Assignment_02.java

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>java Q_06
Enter Your Name :-
Enter Your Age :-
Enter Your Name :- Enter Your Age :-
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>
```

```
File Edit Format View Help
class Q_06{
    public static void main(String args[]){
        System.out.println("Enter Your Name :- ");
        System.out.println("Enter Your Age :- ");
        System.out.println("Enter Your Name :- ");
        System.out.println("Enter Your Age :- ");
    }
}
```

Println() command move the new line and next output open the new line. But print command not change the line. Same line print.

Question 07

```
Enter Your Name :- Enter Your Age :-
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>javac Assignment_02.java

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>java Q_07
a
a

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>_
```

```
File Edit Format View Help
class Q_07{
    public static void main(String args[]){
        System.out.println("a"); //line 1
        System.out.println('a'); //line 2
    }
}
```

The output is same but line 1 is string literal and line 2 is char literal.

Question 08

```
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>javac Assignment_02.java
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>java Q_08
7.0
7
7
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>_

class Q_08{
    public static void main(String args[]){
        System.out.println(7); //line 1
        System.out.println(7.0); //line 2
        System.out.println("7"); //line 3
        System.out.println('7'); //line 4
    }
}
```

Line integer literal

Floating-point literal

String Object

Character literal

Question 09

Intiger Literals

Floating Point Literals

Character Literals

String Literals

Question 10

```
7.0
1100100
100
294976
17826048
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>_

class Q_10{
    public static void main(String args[]){
        System.out.println(1100100); //line 1
        System.out.println(0B1100100); //line 2
        System.out.println(01100100); //line 3
        System.out.println(0x1100100); //line 4
    }
}
```

Line 1 – Decimal Literal

Line 2 – Binary Literal

Line 3 – Octal Literal

Line 4 – Hexadecimal Literal

Question 11

Line 1

Line 2

Line 4

Line 6

Line 7

Line 10

Question 12

```
Exception in thread "main" java.util.NoSuchElementException
    at java.base/java.util.Scanner.throwFor(Scanner.java:975)
    at java.base/java.util.Scanner.next(Scanner.java:1632)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2297)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2251)
    at Example.main(Example.java:8)
^C
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>javac Example.java

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>java Example
Exception in thread "main" java.util.NoSuchElementException
    at java.base/java.util.Scanner.throwFor(Scanner.java:975)
    at java.base/java.util.Scanner.next(Scanner.java:1632)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2297)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2251)
    at Example.main(Example.java:8)
^C
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>javac Example.java

D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>java Example
Exception in thread "main" java.util.NoSuchElementException
    at java.base/java.util.Scanner.throwFor(Scanner.java:975)
    at java.base/java.util.Scanner.next(Scanner.java:1632)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2297)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2251)
    at Example.main(Example.java:8)
^C
D:\ICET\Programming Fundamanel\Programming Fundamantel 01.03.2026>
```

```
class Example{
    public static void main(String args[]){
        System.out.print("A\nB");
        System.out.println("CD");
        System.out.print("\nEF\n");
        System.out.print("G\nH")
    }
}
```

Compile error (line 8 ;)

Question 13

```
Microsoft Windows [Version 10.0.19045.7058]
(c) Microsoft Corporation. All rights reserved.

D:\ICET\Programming Fundamanel\Assignment>notepad Assignment_02.java

D:\ICET\Programming Fundamanel\Assignment>javac Assignment_02.java

D:\ICET\Programming Fundamanel\Assignment>java Assignment_02
ICM ? ICET CERTIFIED MASTER

ICM ICET CERTIFIED MASTER

ICM ICET CERTIFIED MASTER
D:\ICET\Programming Fundamanel\Assignment>
```

```
File Edit Format View Help
class Assignment_02 {
    public static void main(String args[]) {

        System.out.print("ICM - ICET CERTIFIED MASTER\n\n");

        System.out.print("ICM ICET CERTIFIED MASTER\n\n");

        System.out.print("ICM ICET CERTIFIED MASTER");

    }
}
```

Question 14

```
ICM ICET CERTIFIED MASTER
D:\ICET\Programming Fundamanel\Assignment>javac Assignment_02.java

D:\ICET\Programming Fundamanel\Assignment>java Assignment_02
Hello
JAVA
Hello JAVA
Hello JAVA
Hello JAVA\
Hello
JAVA"
Hello
JAVA'

D:\ICET\Programming Fundamanel\Assignment>
```

```
File Edit Format View Help
class Assignment_02 {
    public static void main(String args[]){
        System.out.println("Hello\nJAVA");
        System.out.println("Hello\tJAVA");
        System.out.println("Hello\bJAVA");
        System.out.println("\Hello JAVA\");
        System.out.println("\Hello\nAVA\");
        System.out.println("\Hello\nAVA\");
    }
}
```

Question 15

```
D:\ICET\Programming Fundamanel\Assignment>javac Assignment_02.java
D:\ICET\Programming Fundamanel\Assignment>java Assignment_02
a. Java is a typed language
b. AB*CB
c. AB\CD
d. C:\Windows\Program
e. AB\tCD
f. AB\t*CD
g. AB\nCD
h. AB\tCD
i. AB\bCD
D:\ICET\Programming Fundamanel\Assignment>
```

```
File Edit Format View Help
class Assignment_02 {
    public static void main(String args[]) {
        System.out.println(
            "a. Java is a typed language\n" +
            "b. AB\tCB\n" +
            "c. AB\tCD\n" +
            "d. C:\\Windows\\Program\n" +
            "e. AB\\tCD\n" +
            "f. AB\\t*CD\n" +
            "g. AB\\nCD\n" +
            "h. AB\\tCD\n" +
            "i. AB\\bCD"
        );
    }
}
```

Question 16

Question 17

Question 18

Line 01 – variable declara

Line 02 – variable initialization

Line 01 – Print statement

Question 19

```
D:\ICET\Programming Fundamanel\Assignment>javac Assignment_02.java
D:\ICET\Programming Fundamanel\Assignment>java Assignment_02
102 103
103 102
D:\ICET\Programming Fundamanel\Assignment>
```

```
File Edit Format View Help
class Assignment_02 {
    public static void main(String[] args) {
        int x = 102;
        int y = 103;

        System.out.println(x + " " + y);

        System.out.println(y + " " + x);
    }
}
```

Question 20

Compile time error

Question 21

```
D:\ICET\Programming Fundamanel\Assignment>javac Assignment_02.java
Assignment_02.java:7: error: variable y is already defined in method main(String[])
    int y=200;
    ^
1 error
D:\ICET\Programming Fundamanel\Assignment>
```

```
File Edit Format View Help
class Assignment_02 {
    public static void main(String args[]){
        int x=10,y=20,z=30;
        System.out.println(x);
        System.out.println(y);
        System.out.println(z);
        int y=200;
        System.out.println(y);
    }
}
```

Compile error (the same variables have been declared twice)

Question 22

```
D:\ICET\Programming Fundamanel\Assignment>javac Assignment_02.java
D:\ICET\Programming Fundamanel\Assignment>java Assignment_02
100
200
300
D:\ICET\Programming Fundamanel\Assignment>_
```

```
File Edit Format View Help
class Assignment_02 {
    public static void main(String args[]){
        int x,y;
        x=100;
        y=200;
        System.out.println(x);
        System.out.println(y);
        y=300;
        System.out.println(x);
        System.out.println(y);
    }
}
```

When two values are assigned to the same variable, the sum of the values is printed

Question 23

B,C,D

Question 24

```
100
300
D:\ICET\Programming Fundamanel\Assignment>javac Assignment_02.java
Assignment_02.java:8: error: cannot find symbol
    int i=j=k=10; //line 6
           ^
symbol:   variable j
location: class Assignment_02
Assignment_02.java:8: error: cannot find symbol
    int i=j=k=10; //line 6
           ^
symbol:   variable k
location: class Assignment_02
2 errors
D:\ICET\Programming Fundamanel\Assignment>_
```

```
File Edit Format View Help
class Assignment_02 {
    public static void main( String args[] ){
        int x,y,z; //line 1
        x=y=z=10; //line 2
        int a=12,b,c=4; //line 3
        int p=10; //line 4
        int q=p; //line 5
        int i=j=k=10; //line 6
    }
}
```

Compile error at line 06

Question 25

1020

30

Line 1: "10" + "20" → string concatenation → "1020".

Line 2: 10 + 20 → integer addition → 30.

Question 26

```
D:\ICET\Programming Fundamanel\Assignment>javac Assignment_02.java
D:\ICET\Programming Fundamanel\Assignment>java Assignment_02
60
10+20+30
10+2030
102030
102030
3030
102030
D:\ICET\Programming Fundamanel\Assignment>_
```

```
File Edit Format View Help
class Assignment_02 {
    public static void main(String args[]){
        System.out.println(10+20+30);
        System.out.println("10+20+30");
        System.out.println("10+20"+30);
        System.out.println("10"+"20"+"30");
        System.out.println("10"+20+30);
        System.out.println(10+20+30);
    }
}
```

Question 27

Pure numbers/chars → arithmetic with ASCII values.

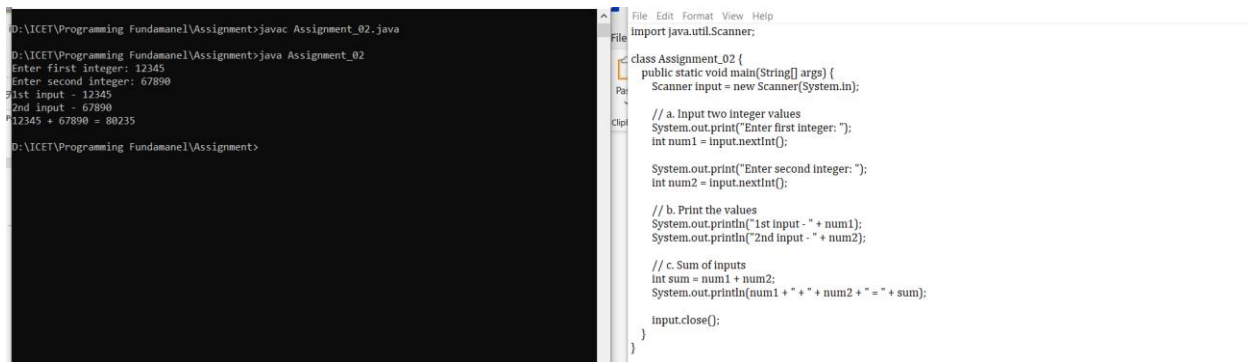
Any string involved → everything becomes string concatenation.

Question 28

Question 20 program → prints fixed results of expressions (like 10+20+30), output is always the same.

This program → takes **user input** at runtime using Scanner, then prints what the user types. Output changes depending on the entered value.

Question 29



```
0:\ICET\Programming Fundamane1\Assignment>javac Assignment_02.java
0:\ICET\Programming Fundamane1\Assignment>java Assignment_02
Enter first integer: 12345
Enter second integer: 67890
1st input - 12345
2nd input - 67890
12345 + 67890 = 80235
0:\ICET\Programming Fundamane1\Assignment>
```

```
File Edit Format View Help
import java.util.Scanner;

class Assignment_02 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        // a. Input two integer values
        System.out.print("Enter first integer: ");
        int num1 = input.nextInt();

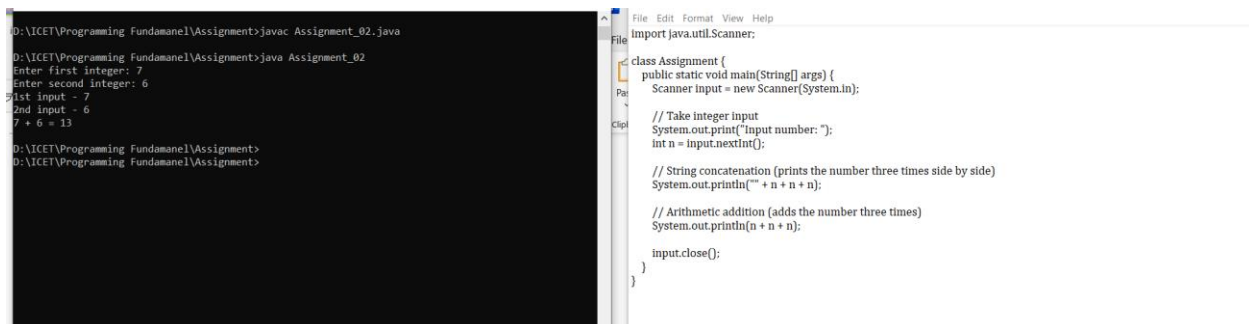
        System.out.print("Enter second integer: ");
        int num2 = input.nextInt();

        // b. Print the values
        System.out.println("1st input - " + num1);
        System.out.println("2nd input - " + num2);

        // c. Sum of inputs
        int sum = num1 + num2;
        System.out.println(num1 + " + " + num2 + " = " + sum);

        input.close();
    }
}
```

Question 30



```
0:\ICET\Programming Fundamane1\Assignment>javac Assignment_02.java
0:\ICET\Programming Fundamane1\Assignment>java Assignment_02
Enter first integer: 7
Enter second integer: 6
1st input - 7
2nd input - 6
7 + 6 = 13
0:\ICET\Programming Fundamane1\Assignment>
0:\ICET\Programming Fundamane1\Assignment>
```

```
File Edit Format View Help
import java.util.Scanner;

class Assignment {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        // Take Integer input
        System.out.print("Input number: ");
        int n = input.nextInt();

        // String concatenation (prints the number three times side by side)
        System.out.println(" " + n + n + n);

        // Arithmetic addition (adds the number three times)
        System.out.println(n + n + n);

        input.close();
    }
}
```

Question 31

```

Total          - 315

D:\ICET\Programming Fundamane1\Assignment>javac Assignment.java

D:\ICET\Programming Fundamane1\Assignment>java Assignment
Combined Maths - 81
Chemistry      - 72
Physics        - 84
English        - 78
Total          - 315

D:\ICET\Programming Fundamane1\Assignment>

```

```

File Edit Format View Help
class Assignment{
    public static void main(String[] args) {
        // Declare and initialize marks
        int combinedMaths = 81;
        int chemistry = 72;
        int physics = 84;
        int english = 78;

        // Calculate total
        int total = combinedMaths + chemistry + physics + english;

        // Print output in required format
        System.out.println("Combined Maths - " + combinedMaths);
        System.out.println("Chemistry      - " + chemistry);
        System.out.println("Physics        - " + physics);
        System.out.println("English        - " + english);
        System.out.println("Total          - " + total);
    }
}

```

Question 32

```

D:\ICET\Programming Fundamane1\Assignment>javac Assignment.java

D:\ICET\Programming Fundamane1\Assignment>java Assignment
Enter Red value : 12
Enter Green value : 75
Enter Blue value : 18
Inversion of given colour - [243, 180, 237]

D:\ICET\Programming Fundamane1\Assignment>

```

```

File Edit Format View Help
import java.util.Scanner;

public class Assignment {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input values
        System.out.print("Enter Red value : ");
        int red = sc.nextInt();

        System.out.print("Enter Green value : ");
        int green = sc.nextInt();

        System.out.print("Enter Blue value : ");
        int blue = sc.nextInt();

        // Invert values (255 - value)
        int invertedRed = 255 - red;
        int invertedGreen = 255 - green;
        int invertedBlue = 255 - blue;

        // Output
        System.out.println("Inversion of given colour - ["
            + invertedRed + ", "
            + invertedGreen + ", "
            + invertedBlue + "]");

        sc.close();
    }
}

```

Question 33

Data Type	Size	Description
byte	1 byte	-128 to 127
short	2 bytes	-32766 to 32767
int	4 bytes	-2147483648 to 2147483647
long	8 bytes	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
float	4 bytes	-precision 32-bit IEEE 754 floating point
double	8 bytes	-precision 64-bit IEEE 754 floating point
boolean	1 bit	-true or false
char	2 bytes	-Single 16-bit Unicode character

Question 34

@name

new

user-input

My Variable

Boolean

123num

Java.org

Question 35

Line 1

A fixed string literal "iCET"

Always prints the same text

Direct output

Line 2

The value stored in variable s

Can print different values if s changes

Uses a variable to hold and print data

Question 36

```
D:\ICET\Programming Fundamanel\Assignment>javac Assignment.java
D:\ICET\Programming Fundamanel\Assignment>java Assignment
Enter your name: Github
Hello Github, nice to meet you!
D:\ICET\Programming Fundamanel\Assignment>
```

```
File Edit Format View Help
import java.util.Scanner;

class Assignment {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Ask for the user's name
        System.out.print("Enter your name: ");
        String name = sc.nextLine();

        // Display greeting
        System.out.println("Hello " + name + ", nice to meet you!");

        sc.close();
    }
}
```

Question 37

```
D:\ICET\Programming Fundamanel\Assignment>javac Assignment.java
D:\ICET\Programming Fundamanel\Assignment>java Assignment
z is: 131y105z
D:\ICET\Programming Fundamanel\Assignment>
```

```
File Edit Format View Help
class Assignment {
    public static void main(String[] args) {
        int x = 10;
        int y = 5;
        String z = x + 'y' + 'y' + x + y + 'z';
        System.out.println("z is: " + z);
    }
}
```

The output does not match the answers given

Question 38

```
D:\ICET\Programming Fundamane1\Assignment>javac Assignment.java
```

```
D:\ICET\Programming Fundamane1\Assignment>java Assignment
```

```
Enter your friend's name: Nipuni  
Enter your friend's age: 23  
Enter your friend's living place: kaluthara
```

```
Nipuni is my best friend. He is 23 years old and lives in the beautiful town of kaluthara.
```

```
D:\ICET\Programming Fundamane1\Assignment>
```

File Edit Format View Help

```
import java.util.Scanner;
```

```
class Assignment {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        // Input friend's details  
        System.out.print("Enter your friend's name: ");  
        String name = sc.nextLine();  
  
        System.out.print("Enter your friend's age: ");  
        int age = sc.nextInt();  
  
        sc.nextLine(); // consume newline  
  
        System.out.print("Enter your friend's living place: ");  
        String place = sc.nextLine();  
  
        // Output description  
        System.out.println();  
        System.out.println(name + " is my best friend. He is "  
            + age + " years old and lives in the beautiful town of "  
            + place + ".");  
  
        sc.close();  
    }  
}
```

Question 39

```
D:\ICET\Programming Fundamane1\Assignment>javac Assignment.java
```

```
D:\ICET\Programming Fundamane1\Assignment>java Assignment
```

```
---Subject total calculator----
```

```
Subject 1 name - Bio
```

```
Bio marks - 80
```

```
Subject 2 name - Maths
```

```
Maths marks - 85
```

```
Subject 3 name - Music
```

```
Music marks - 95
```

```
Bio
```

```
80
```

```
Maths
```

```
85
```

```
Music
```

```
95
```

```
Total
```

```
260
```

```
D:\ICET\Programming Fundamane1\Assignment>
```

File Edit Format View Help

```
import java.util.Scanner;
```

```
class Assignment {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("---Subject total calculator---");  
  
        // Input subject 1  
        System.out.print("Subject 1 name - ");  
        String subject1 = sc.nextLine();  
        System.out.print(subject1 + " marks - ");  
        int marks1 = sc.nextInt();  
        sc.nextLine(); // consume newline  
  
        // Input subject 2  
        System.out.print("Subject 2 name - ");  
        String subject2 = sc.nextLine();  
        System.out.print(subject2 + " marks - ");  
        int marks2 = sc.nextInt();  
        sc.nextLine();  
  
        // Input subject 3  
        System.out.print("Subject 3 name - ");  
        String subject3 = sc.nextLine();  
        System.out.print(subject3 + " marks - ");  
        int marks3 = sc.nextInt();  
  
        // Calculate total  
        int total = marks1 + marks2 + marks3;  
  
        // Output  
        System.out.println();  
        System.out.println(subject1 + "    " + marks1);  
        System.out.println(subject2 + "    " + marks2);  
        System.out.println(subject3 + "    " + marks3);  
        System.out.println();  
        System.out.println("Total    " + total);  
  
        sc.close();  
    }  
}
```

Question 40

```
D:\ICET\Programming Fundamane\Assignment>javac Assignment.java
```

```
D:\ICET\Programming Fundamane\Assignment>java Assignment
```

```
Input number 1 : 15
```

```
Input number 2 : 75
```

```
15 75
```

```
75 15
```

```
D:\ICET\Programming Fundamane\Assignment>
```

```
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```

```
import java.util.*;
```

```
class Assignment {
```

```
    public static void main(String args[]) {
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input number 1 : ");
```

```
        int num1 = input.nextInt();
```

```
        System.out.print("Input number 2 : ");
```

```
        int num2 = input.nextInt();
```

```
        // First output (before swap)
```

```
        System.out.println(num1 + " " + num2);
```

```
        //-----Do not modify before this line-----
```

```
        // Swap logic
```

```
        int temp = num1; // store num1 in a temporary variable
```

```
        num1 = num2; // assign num2 to num1
```

```
        num2 = temp; // assign temp (original num1) to num2
```

```
        //-----Do not modify after this line-----
```

```
        // Second output (after swap)
```

```
        System.out.println(num1 + " " + num2);
```

```
    }
```

```
}
```